SOTA INOUE

Bachelor of Engineering

Personal Information

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Nationality: Japan

Date of Birth: January 9, 2002

Language

- Japanese (Native)
- English (B2+ Independent- Proficient)
- Germany (B1.2, currently learning)

Expertise

- Computational Neuroscience
- Neural Engineering
- Behavioral Decision-Making
- Observational Learning
- Neural Circuit Modeling
- Cognitive & Systems Neuroscience

Professional Summary

Neuroscience graduate with a strong engineering background, specialized in computational modeling, reinforcement learning, and experimental neuroscience.
Experienced in Python-based data analysis, behavioral task design, and interdisciplinary research at the interface of neuroscience and engineering.

Research Experience

Research Assistant (HiWi), Neuropsychology Lab, University of Freiburg
Jun 2026 – Sep 2026
EEG, sleep, memory, and physiological signal analysis; research support for neuropsychological studies.

Independent Research Project, Kyoto University
Apr 2024 – Mar 2025
Computational modeling of self-learning and observational learning using reinforcement learning, inverse reinforcement learning, and behavioral experiments.

Research Experience, ATR Brain Function Institute
2024
Assisted with human fMRI experiments and gained exposure to neuroimaging data acquisition and analysis workflows.

Education

University of Freiburg, Germany
Master of Science in Neuroscience
Oct 2025 – Present

Kyoto University, Japan
Bachelor of Engineering in Electrical and Electronic Engineering
Apr 2021 – Mar 2025
Bachelor Thesis: Computational Modeling of Decision-Making integrating Self-Learning and Observational Learning

Kyoto University, Japan
Master of Informatics (Computational Neuroscience) – on leave
Apr 2025 – Present

Additional Technical Experience

- Web development (TypeScript, Node.js, React)

Hobbies

- Independent travel, endurance training (running, calisthenics),
- football (13 years)
- podcast listening

Teaching & Academic Engagement

Kyoto Consortium for Japanese Studies (KCJS)

Support Instructor (Sep 2022 – Sep 2024)

- Supported international students in Japanese language learning and academic adaptation.

Kyoto University International Summer Program (KSP)

Participant (2022), Organizer (2023)

- Contributed to the organization of an international academic program.

Vietnam National University, Hanoi (ULIS & USSH)

Spring Vietnamese–Japanese Intercultural Program (Feb – Mar 2023)

- Led a student team in an intercultural academic presentation.

Research Methods

Neuroscience & Experimental Methods

- Neural signal processing (membrane to systems level)
- Neuronal signaling and synaptic plasticity
- Sensory and motor systems
- Behavioral and cognitive task design
- Psychophysics experiments (PsychoPy)
- EEG and physiological signal analysis

Computational & Analytical Methods

- Computational modeling of learning and decision-making
- Reinforcement Learning (RL) and Inverse Reinforcement Learning (IRL)
- Neural circuit analysis and conceptual modeling
- Time–frequency analysis of neural signals
- Statistical analysis and data visualization

Technical Skills

- Programming: Python, C
- Scientific computing and data analysis
- Machine learning and reinforcement learning
- Tools: Linux (Ubuntu), Git, Docker, Arduino

Internship

System Engineer Intern

Nocnum (Deep-Tech Startup), Japan — Apr 2024 – Present

- Developed and maintained frontend and backend systems for IoT-based water purification monitoring using TypeScript, Node.js, React, and Docker.
- Implemented sensor-based data acquisition systems, integrating electronic hardware and software components.
- Conducted data analysis on water quality sensor data and led the implementation of a clustering-based analysis pipeline.
- Utilized Python and machine learning methods for unsupervised clustering of sensor data.
- Worked in an international team environment and contributed to system development and deployment using Linux-based environments.

Honors and Awards

DAAD Master's Scholarship (2025–2027)
German Academic Exchange Service (DAAD)

- Highly competitive scholarship awarded by the German Federal Foreign Office for outstanding academic performance and research potential.

TA Award, Electronics Spring Camp (2022 Spring)
Kyoto University, Department of Electrical and Electronic Engineering

- Awarded for outstanding performance in team-based electronic system design using LEGO Mindstorms.

Grand Prize (Iwakabe Special Award), Electronics Summer Camp (2022 Summer)
Kyoto University, Department of Electrical and Electronic Engineering

- Led a team to develop an **Arduino**-based line-tracing robotic system and received the highest award for electronic system design.